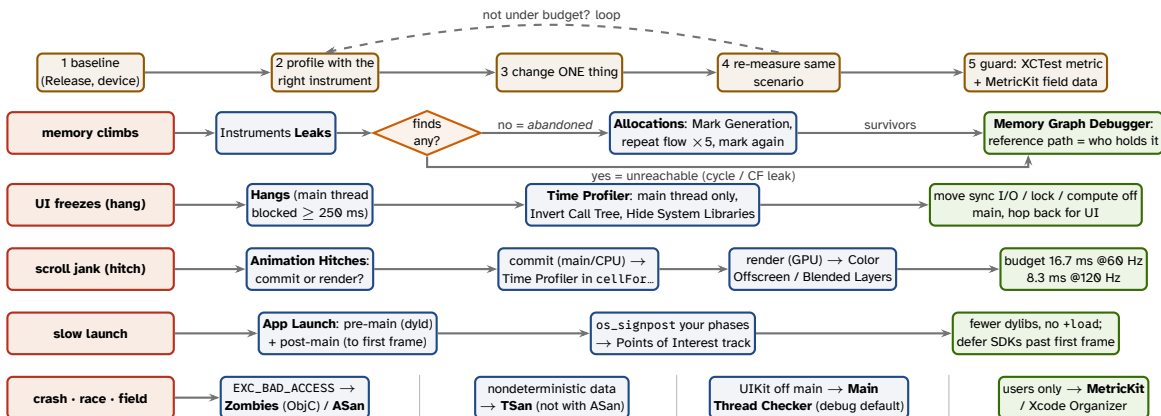


In one line: Measure, don't guess: baseline on a **real device** in the **Release/Profile** config → pick the instrument that answers *this* symptom → change **one** thing → re-measure the same scenario. For each tool know its **name**, how to **read its number**, and what it **cannot** see.



The tools, precisely

- **Memory Graph Debugger** (Xcode debug bar, *not* Instruments): pauses the app, snapshots every live object and its references. Purple ! = leak/cycle; select an object → *who holds it*. Snapshot only, no timeline. Enable **Malloc Stack Logging** (scheme → Diagnostics) to get the *allocation* *backtrace*.
- **Leaks** (Instruments): periodically scans for **unreachable** blocks — cycles, un-released CF. **Blind to abandoned memory** (still reachable: unbounded Dictionary cache, observer array).
- **Allocations**: every heap alloc over time. **Persistent** vs **Transient**; **Mark Generation** (a.k.a. heapshot) — objects created after a mark and still alive show under that generation. Stepwise growth per repetition = abandoned or leaked.
- **Footprint** ≠ **live bytes**: Jetsam kills on *footprint* (dirty pages, decoded bitmaps). 4000×3000 image = 48 MB decoded, any JPEG size.
- **Time Profiler** *samples* every thread's stack (~1 ms): statistical, cheap, no call counts. Read it: *Separate by Thread* → main thread, *Invert Call Tree*, *Hide System Libraries*; chase high **self** weight (total includes callees). 80 % CPU on a background thread is harmless; 300 ms of sync I/O on main is a hang.
- **Hang** = main thread not servicing the run loop (Hangs instrument, MXHangDiagnostic). **Hitch** = frame presented late. Different metric, different fix. **Thread Performance Checker** flags priority inversions at runtime; **System Trace** shows *why* a thread is not running (lock, page fault).
- **Hitch time ratio** = hitch ms per second of scroll: < 5 ms/s good, 5–10 warning, > 10 critical. Commit hitch = your layout/decode on main; render hitch = offscreen passes (masksToBounds+radius, shadow w/o shadowPath), blending.
- **Launch**: pre-main = dyld loads/binds dylbs, ObjC setup, static initializers; post-main = main → didFinishLaunching → first frame. Measure *cold* launch; the watchdog (0x8badf00d) kills a stuck launch. CI: measure(metrics: [XCTApplicationLaunchMetric()]).
- **MetricKit**: MXMetricPayload (launch, hang rate, memory, hitches — aggregated ~daily) + MXDiagnosticPayload (hang/crash/CPU/disk-write stacks). Tells you *which* problem to bring into Instruments.

Example — signpost + MetricKit

```
import os; import MetricKit
let sp = OSSignposter(subsystem: "app.feed",
                      category: .pointsOfInterest)
func loadFeed() async throws -> [Post] {
    let s = sp.beginInterval("Feed", id: sp.makeSignpostID())
    defer { sp.endInterval("Feed", s) } // ~free unless recording
    return try await api.fetchFeed()
}
final class Field: NSObject, MXMetricManagerSubscriber {
    func didReceive(_ p: [MXMetricPayload]) {} // aggregates
    func didReceive(_ p: [MXDiagnosticPayload]) {} // hang stacks
}
MXMetricManager.shared.add(field) // keep 'field' alive
```

Traps

- ! “Leaks says 0, so no leak” — memory still climbs = *abandoned* memory. Allocations generations, then Memory Graph for the holder.
- ! Memory Graph lives in **Xcode**, Leaks/Allocations in **Instruments** — three tools, three questions.
- ! Profiling **Debug** or the **Simulator**: no inlining, Mac CPU — wrong hot spots. Profile scheme, slowest supported device. NSLog timing perturbs the run — use os_signpost.
- ! Dealloc test: XCTAssertNil inside DispatchQueue.main.async asserts *nothing* — the test already returned. Assert directly or await an expectation.
- ! Zombies: ObjC objects only, never frees. ASan + TSan: not in one run.
- ! “16 ms per frame” on a ProMotion phone: it is **8.3 ms**.

Runtime checkers (scheme → Diagnostics)

Address Sanitizer	use-after-free, buffer overflow (C, Swift unsafe)
Thread Sanitizer	data races — not together with ASan
Undefined Behavior Sanitizer	C/ObjC undefined behaviour (overflow, misaligned)
Main Thread Checker	UIKit/AppKit off main — on by default in Debug
Thread Performance Checker	priority inversions, main-thread I/O
Zombie Objects	message to a freed ObjC object

Remember

Growth: **Leaks** → **Allocations** → **Graph**. Freeze: **Hangs** → **Time Profiler** (main, inverted). Jank: **Hitches** → **commit** or **render**. Field: **MetricKit**. Always: *one change, re-measure*.